

Research Article: Traffic Accidents Analysis Using Machine Learning

Traffic Accidents Analysis and Prediction Using Data Analytics

Abstract

Traffic accidents are a major public safety concern worldwide, causing injuries, fatalities, and economic losses. This study analyzes traffic accident data using data visualization and machine learning techniques to identify factors influencing road accidents. Various attributes such as weather conditions, lighting conditions, crash type, trafficway type, day of the week, and crash hour are examined to discover patterns and trends. A Decision Tree Classification model is applied to predict accident-related outcomes and support road safety improvements.

Introduction

Road traffic accidents result in significant human and financial losses every year. Understanding the causes and patterns of accidents can help governments and transportation authorities develop effective safety measures. Data analytics and machine learning provide powerful tools for exploring accident datasets and predicting accident risks.

The objective of this project is to:

- Analyze traffic accident data.
- Identify factors contributing to accidents.
- Visualize accident trends and patterns.
- Build a predictive model using machine learning.

Dataset Description

The dataset contains information related to traffic accidents, including:

- Crash Date
- Crash Month
- Crash Hour
- Day of Week
- Weather Condition
- Lighting Condition

- Trafficway Type
- Crash Type
- First Crash Type
- Fatal Injuries
- Incapacitating Injuries
- Non-incapacitating Injuries
- No Injury Cases

Methodology

1. Data Preprocessing

- Loaded dataset using Pandas.
- Checked missing values.
- Generated statistical summaries.
- Converted categorical data into numerical format using Label Encoding.

2. Exploratory Data Analysis (EDA)

Several visualizations were created:

Correlation Heatmap

Used to identify relationships among numerical variables.

Monthly Accident Analysis

Shows accident frequency across different months.

Lighting Condition Analysis

Examines how lighting conditions affect accident occurrence.

Weather Condition Analysis

Studies accident patterns under different weather conditions.

Crash Type Analysis

Identifies the most common categories of accidents.

Hourly Accident Distribution

Analyzes accident frequency during different times of the day.

Day-of-Week Analysis

Determines which days experience the highest accident rates.

Injury Analysis

Compares different injury severity levels resulting from accidents.

3. Machine Learning Model

A Decision Tree Classifier was used because:

- Easy to understand and interpret.
- Suitable for classification tasks.
- Handles categorical and numerical features effectively.

The dataset was divided into:

- Training Set (80%)
- Testing Set (20%)

Results and Findings

Key Observations

1. Accident frequency varies significantly by month.
2. Poor lighting conditions contribute to higher accident rates.
3. Weather conditions influence accident occurrence.

4. Certain crash types occur more frequently than others.
5. Peak accident hours are usually during busy traffic periods.
6. Injury severity depends on crash type and road conditions.

Machine Learning Performance

The Decision Tree model successfully learned patterns from the accident dataset and provided predictions based on accident-related features. The model can assist in identifying high-risk situations and improving traffic management strategies.

Conclusion

Traffic accident analysis helps identify critical factors responsible for road crashes. Through data visualization and machine learning techniques, valuable insights can be extracted to support road safety planning. The Decision Tree model demonstrates the potential of predictive analytics in reducing accident risks and improving transportation safety.

Future Work

- Use larger and more recent datasets.
- Apply advanced algorithms such as Random Forest and XGBoost.
- Develop real-time accident prediction systems.
- Integrate GIS mapping for accident hotspot analysis.

References

1. World Health Organization
2. Pandas
3. Scikit-learn
4. Matplotlib
5. Seaborn